

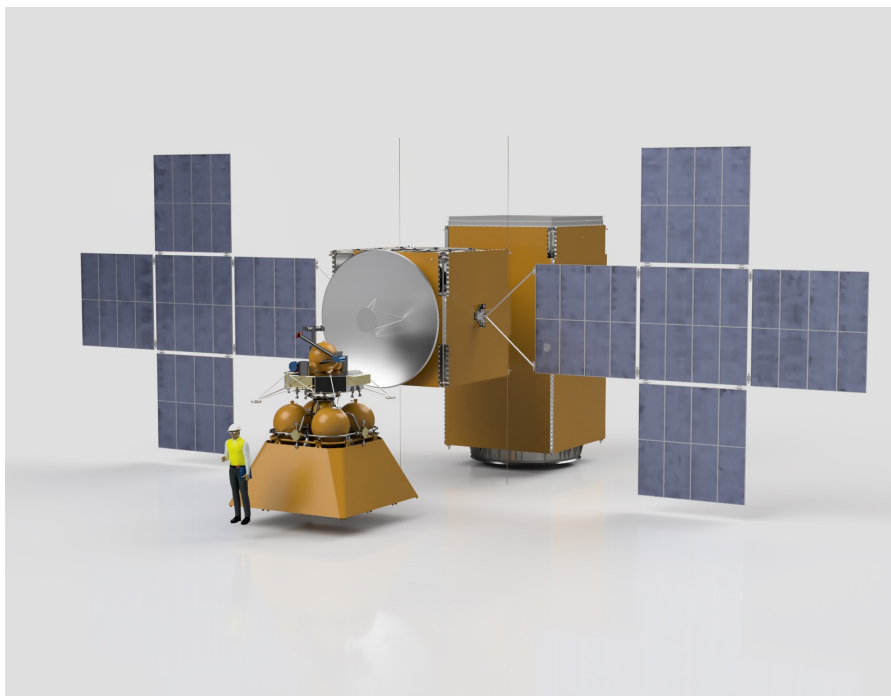
AERO60004 - AERONAUTICS GROUP DESIGN PROJECT

IMPERIAL COLLEGE LONDON

DEPARTMENT OF AERONAUTICS

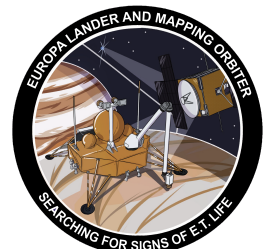
Europa Lander and Mapping Orbiter Lander Team - Scientific Instruments and Operations Design

Supervisor: Dr. Gustavo Quino Quispe



Arthur Law

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Subteams and Members

Project Coordination	Jack Osborn Victor Djololian Charles Kempf
-----------------------------	--

Orbital Trajectory	Egor Kaygorodov Debdut Sengupta Owain Evans Yili Chen
---------------------------	--

Transfer Vehicle	Parth Gore Sriram Iyer Murray Doolan Jack Crane Léa Ruas
-------------------------	--

Lander	Klim Kapuka Miharu Aleks Kent Oliver Jackson Patrick Fan Jason Aretaois Hong Arthur Law
---------------	---

EDL	Jacob Dodridge Hengbo Cosmo Zhou Jirapat Rojnuckarin
------------	--

Communications	Andrei-Dinca Paduraru Lewis Mossaberi Ewan Sinclair Gaétan Bouscaillou
-----------------------	---

Abstract

This report presents the design and analysis of a mission concept for a Europa lander aimed at searching for biomarkers and signs of life. Europa, one of Jupiter’s moons, is a prime candidate for extraterrestrial life due to its subsurface ocean, which is hypothesized to be rich in minerals and potentially in contact with a rocky mantle, similar to Earth’s deep-ocean hydrothermal ecosystems. The mission objectives include landing on Europa, drilling 10 cm into the surface to extract and analyse samples, and using a suite of instruments to assess habitability and surface properties to support future exploration.

A preliminary analysis was conducted, reviewing previous missions such as Viking, Curiosity, and Cassini, and ongoing research including Instrument Concepts for Europa Exploration 2 (ICEE-2). The modal payload includes:

1. Organic Compositional Analyser (OCA): Capable of detecting organic molecules with a detection limit of 0.01 parts per billion by weight and a mass range of 2-535 Daltons.
2. Vibrational Spectrometer (VS) and Microscope: Using deep ultraviolet (DUV) laser-induced fluorescence and Raman spectroscopy to detect molecular compositions, with spatial resolutions of 10.1 micrometers and 15.9 micrometers for the autofocus and WATSON cameras, respectively.
3. Color Stereo Imagers: Providing high-definition, 3D imaging with resolutions of 450 microns per pixel at 2 meters for Mastcam-34 and 150 microns per pixel for Mastcam-100.
4. Geophysical Sounding System (GSS): Utilising short-period seismometers to measure ice shell and ocean thickness with a frequency range of 0.1 to 100 Hz.

The mission design ensures all instruments fit within a 650 x 600 x 280 mm aluminum vault with a total volume constraint of 95 liters and a payload mass of 70 kg. Data transmission capabilities are set to 9 Gbits over the mission duration, with a power consumption of 1117.5 Whr per sampling cycle. The mission timeline includes four sampling tals over a 20-day period, allowing for additional trenching and analysis opportunities, ensuring comprehensive data collection and transmission.

This report provides a comprehensive overview of the mission’s feasibility, design choices, and operational strategies, highlighting the scientific potential of exploring Europa’s subsurface ocean and contributing to the broader search for life beyond Earth.

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1 Introduction

On Earth, life thrives wherever water is present. Thus, Europa, with its subsurface ocean, is a prime candidate in the search for extraterrestrial life. This ocean, potentially warmed by geological activity and enriched with minerals and organic compounds, might support life. Various Earth ecosystems rely on hydrothermal vents and chemosynthesis, bypassing sunlight altogether. Measurements suggest Europa’s ocean is twice the volume of Earth’s oceans [23], possibly in contact with the moon’s interior, providing access to hydrothermal energy. Europa’s active surface may regularly bring subsurface materials to the surface, aiding exploration.

The Group Design Project aims to design a Europa lander mission to:

1. Land on Europa, drill 10 cm, extract, and analyse a sample to search for biomarkers or signs of life,
2. Use a microscope, camera, and seismometer to assess Europa’s habitability and characterise surface properties to support future exploration,
3. Achieve a success probability over 95%,
4. Remain operational for at least 1 month on Europa.
5. Perform a hopping maneuver to another site to collect additional samples.

This report primarily aims to explore the operational standards required to maximise the overall chance of mission success. Additionally, it outlines an optimal selection of payload functionality and necessary design modifications to ensure that volume, mass, and data budget constraints are met for this project.

1.0.1 Guidance / Definitions / Methodology

The 2016 Europa Lander Science Definition Team Study Report [33] dives deep into considerations of scientific instruments and operations with industrial data to support design choices. The NASA Astrobiology Roadmap [11] offers extensive definitions of signatures of life and habitable environments on other worlds. The Mission Analysis Hierarchy from SMAD [34] provides guidelines for evaluating mission success.

Analysis Type	Goal	
Feasibility Assessment	To establish whether an objective is achievable and its approximate degree of complexity	Quick, limited detail
Sizing Estimate	To estimate basic parameters such as size, weight, power or cost	
Point Design	To demonstrate feasibility and establish a baseline for comparison of alternatives	
Trade Study	To establish the relative advantages of alternative approaches or options	More detailed, complex trades
Performance Assessment	To quantify performance parameters (e.g., resolution, timeliness) for a particular approach	
Utility Assessment	To quantify how well the system can meet overall mission objectives	

Table 1: Mission Analysis Hierarchy from SMAD

2 Preliminary Analysis

2.1 Feasibility Assessment

2.1.1 Previous Missions

Viking 2 landed on Mars in September 1976 , following Viking 1, as part of NASA’s mission to investigate Mars and search for life. Although neither spacecraft found definitive life traces, they detected essential elements: carbon, nitrogen, hydrogen, oxygen, and phosphorus [19]. The Mars Science Laboratory’s Curiosity rover

(2012) uses the Sample Analysis at Mars (SAM) suite to analyse organics and gases, and the Chemistry and Mineralogy (CheMin) instrument to identify water-formed minerals [21]. Curiosity detected complex organic molecules in Gale Crater [15], suggesting ancient Mars had ingredients for life. The Mars 2020 Perseverance rover includes SHERLOC to detect organics and minerals and MOXIE to produce oxygen from Martian CO₂, crucial for future human missions [24]. Beyond Mars, the Cassini mission to Saturn’s moon Enceladus discovered water-ice geysers, indicating a subsurface ocean with hydrothermal activity [20].

2.2 Ongoing Research

Instrument Concepts for Europa Exploration 2 (ICEE-2) supports the development of instruments and sample transfer mechanisms specifically designed to tackle the challenges of Europa surface exploration, aiming to advance technical readiness for a potential lander mission [26]. Fourteen proposals cover various classes in the Science Definition Team report, including a sample handling instrument and two additional instruments [29]. However, while ICEE-2 instruments seem promising, they are not fully established. Therefore, it is recommended that they are avoided from this project’s payload design. Instead, instruments from previous missions offer reliable specifications and performance data, aiding in accurate success rate estimates. Previous mission instruments also provide a tested operational period, ensuring mission timeline feasibility.

PI	Title	Instrument class	Institution
Byrne	Cold-Lightweight Imagers for Europa (C-LIFE)	CRSI	U. of Arizona
Bailey	Seismometer to Investigate Ice and Ocean Structure (SIOS)	GSS	U. of Arizona
Murchie	Europa Lander Stereo Spectral Imaging Experiment (ELSSIE)	CRSI	JHU/APL
Panning	Europa Seismic Package	GSS	JPL
Grimm	Europa Magnetotelluric Sounder (EMS)	GSS	SWRI
Quinn	Europa Luminescence Microscope	MLD	NASA ARC
Arevalo	CORALS: Characterization of Ocean Residues and Life Signatures	OCA	U. of Maryland
Brinckerhoff	European Molecular Indicators of Life Investigation (EMILI)	OCA	NASA GSFC
Glein	MASS Spectrometer for Planetary EXploration-ORGanic Composition Analyzer (MASPEX-ORCA) for Europa Lander	OCA	SWRI
Mathies	Microfluidic Organic Analyzer for Biosignatures (MOAB)	OCA	UC-Berkeley
Lambert	Compact Integrated Raman Spectrometer (CIRS)	VS	JPL
Malespin	Collaborative Acceptance and Distribution for Measuring European Samples (CADMES) System	Sampling system	NASA GSFC
Moldwin	Instrument Concepts for Europa Exploration (ICEE-2): Reduced SWAP +C Radiation Tolerant Magnetometer for Europa Lander	Magnetometer on a chip	U. of Michigan
Ricco	MICA: Microfluidic Icy-World Chemistry Analyzer	Wet inorganic chemistry	NASA ARC

Figure 1: ICEE-2 Proposals

This list provides a clear understanding of the ideal payload configuration and emphasises design considerations necessary to address the specific challenges posed by Europa’s surface.

2.2.1 Expected Biomes and Organisms

Analysing expected environments on Europa prepares us for specifying scientific instruments. Two types of environments relevant to Europa are:

1) Subsurface Oceans The most promising biome on Europa is its subsurface ocean, estimated to be in contact with a rocky mantle. This environment might be analogous to Earth’s deep-ocean hydrothermal vent ecosystems, where chemosynthetic organisms, thrive despite the absence of sunlight. Key features of Europa’s oceanic biome include:

- **Hydrothermal Vent Systems:** On Earth, hydrothermal vents host diverse ecosystems powered by chemosynthesis, where microorganisms, such as Nitrososphaerota, derive energy from chemical reactions rather than sunlight. Similar vents on Europa could support life forms that use hydrogen sulfide or methane as energy sources. For example, methanogens, a type of archaea, could thrive in such environments by producing methane from carbon dioxide and hydrogen. These microorganisms, along with small plankton like Pelagibacterales and Prochlorococcus [3], guide the resolution needed to detect cells on Europa, **down to 200 nm**.

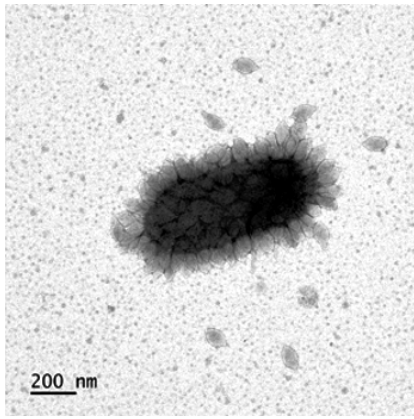


Figure 2: Thaspiviridae, species Nitmarvirus NSV1

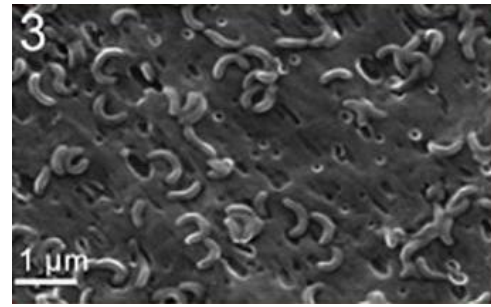


Figure 3: SAR11, a type of Pelagibacteriales [8]

- **Ice-Water Interface:** The dynamic boundary where Europa's ice shell meets the liquid ocean below could harbor microbial life. Extremophiles, an organism that is able to live (or in some cases thrive) in extreme environments, similar to those found in Antarctic subglacial lakes accretion ice, e.g, Lake Vostok [4], might exist here. These microbes could exploit nutrients and chemical gradients created by the freezing and melting processes. For instance, psychrophilic (cold-loving) bacteria and archaea might be found, metabolising organic molecules trapped in the ice. By definition, a detection threshold of **100 cells per mL** is established to match the level of detection on Earth.

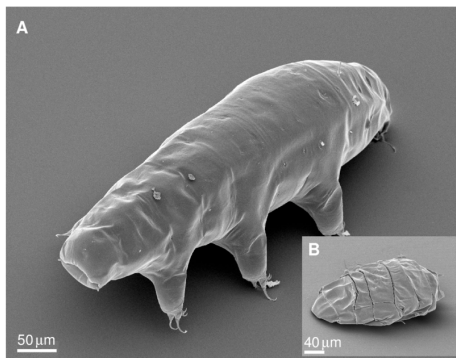


Figure 4: Milnesium tardigradum, a type of Extremophile [28]

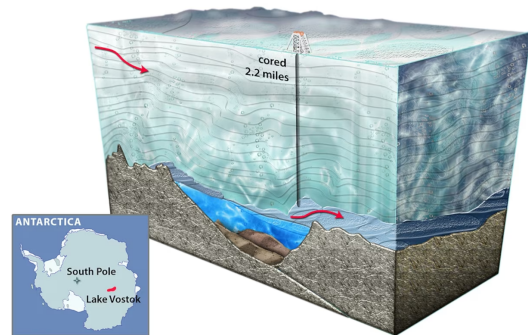


Figure 5: A cross-section of Lake Vostok, the largest known subglacial lake in Antarctica [7]

2) Ice Shell While the ice shell itself is less likely to support complex life, it could still contain microbial ecosystems.

- **Surface Ice:** Jupiter's intense radiation could drive chemical reactions on the surface, creating oxidants like hydrogen peroxide and oxygen. If these oxidants are transported to the ocean below, they could serve as energy sources for microbial life. On Earth, microbes known as radiotrophs, which utilise radiation for metabolic processes, could be analogs for potential European life forms [32].
- **Cryoconite Holes:** These small meltwater pockets within the ice could create isolated habitats where life might persist. On Earth, cryoconite holes on glaciers host microbial communities that survive extreme conditions. Similarly, European cryoconite holes could contain extremophiles such as *Deinococcus radiodurans*, known for their resistance to radiation [31].

	Lake Vostok (Subglacial)			Lake Vida (Salty)		Winter Circumpolar Deep Water (Deep Ocean) ⁴
	Accretion Ice (Type I) ¹	Accretion Ice (Type II) ¹	Glacial Ice ¹	Brine ^{2,3}	Ice ³	
Organic carbon (μM)	65	35	16	64,700	n.a.	41 $\pm$ 3
DFAA (nM); DFAA % Org. Carbon	1-45; $\leq$ 0.006-0.17%	50-174; 0.08-0.49%	20-62; 0.6 – 1.2 %	n.a.*	n.a.	88 $\pm$ 16 ; 0.7 $\pm$ 0.1 %
Total Asp (nM)	15-49	8	11-39	n.a.*	n.a.	n.a.
DF L-Asp (nM)	6-10 ^a	n.d.	10 ^{aa}	n.a.*	n.a.	3-9 [#]
Cell density (cells mL ⁻¹)	260	80	120	49,000,000	444,000	30,000 to 100,000
Microbial size (μm)	$\sim$ 0.3 - 3.0	$\sim$ 0.3 - 3.0	$\sim$ 0.3 - 3.0	0.1-1	$\sim$ 0.5 - 2	0.2 – 1

Figure 6: Cell abundance, densities and size ranges at Earth environments [4][13][14][18][30][33]

*DFAA stands for dissolved free amino acids, and Asp stands for spartic acid and aspartate

2.2.2 Potential Bio-signatures

Bio-signatures can indicate life and chance of habitability.

1) Organic Molecules Organic molecules indicate biological activity on Europa.

- Complex Organic Compounds: Amino acids, lipids, and nucleic acids are fundamental to life. For example, the presence of amino acids like glycine or alanine, especially in chiral forms that favor one orientation (homochirality), could be indicative of biological processes [31].
- Biogenic Gases: Methane (CH_4) is a key bio-signature gas. On Earth, methanogenic archaea produce methane as a metabolic byproduct. Detecting elevated levels of methane, especially with isotopic ratios that suggest a biological origin, would be significant. Additionally, other gases like ethane (C_2H_6) or propane (C_3H_8) in specific proportions could indicate biogenic activity.

2) Morphological Indicators Physical structures left by microorganisms could provide evidence of life.

- Microfossils: Fossilised remains of microorganisms could be preserved in the ice or sediments. Features such as filaments, spheres, or other cell-like shapes could be indicative of past microbial life.
- Biofilms: Microbial communities often form biofilms, which are layers of cells adhering to surfaces. These biofilms can create distinct patterns and textures. For instance, microbial mats found in hot springs and other extreme environments on Earth could serve as analogs.

3) Chemical Signatures Chemical imbalances and specific elemental ratios indicates life processes.

- Isotopic Ratios: Life preferentially uses lighter isotopes, resulting in distinctive isotopic signatures. For example, a high ratio of carbon-12 to carbon-13 (^{13}C) could suggest biological activity, as organisms preferentially use the lighter isotope.
- Redox Gradients: Life relies on redox reactions, which involve the transfer of electrons. Sharp chemical gradients, such as those involving oxygen (O_2), sulfur (H_2S), or nitrogen compounds (NO_3), could indicate metabolic processes. For instance, the presence of sulfates or nitrates in concentrations different from those expected by abiotic processes could point to biological activity.

2.3 Sizing Estimate and Requirements

Table 2 states the constraints given by Communications team and the rest of the Lander team. Communications to the orbiter will be conducted through the use of a UHF quadrifilar helix antenna use in Mars

proximity links, which has a 9 Gbit transfer/downlink constraint over the mission duration [1]. Scientific instruments will be stored within a 650 x 600 x 280mm Aluminium vault with a 10mm wall thickness for radiation shielding, giving a volume constraint of 95 L (including computing systems to be stored). 70 kg is set as the initial allocated payload mass budget from the total lander mass off of SMAD guidelines [34].

Table 3 states the requirements for operating environments for handling, transportation, and storage, according to the Europa Lander Project Preliminary Environmental Requirements Document [22]. These values are passed on to the thermal team to ensure the local lander radiation environment would be mitigated with a combination of hardened technology and shielding, making sure all electronics within the vault are to be rated to 300 krad in order to maintain a radiation design factor of two ($RDF = 2$) [33].

Parameter	Value
Science payload mass (with margin)	70 kg
Inner vault volume	95 L
Minimum number of samples to be collected	1 per site
Required sampling depth capability	10+ cm
Limit of detection for organics	100 cells per mL
Mission Duration	30+ days
Radiation shielding	Central vault
Data volume capability (Total)	Less than 9 Gbits
Total Instrument Energy	2125 Whrs per Europa Day
Instrument command cycles (within 1 orbit)	22 days
Earliest launch date	2026-2027

Table 2: Sizing Estimate for Lander Payloads

Control Parameter	Low Limit	High Limit
Air Temperature (Storage)	+5°C	+50°C
Air Temperature (Operational)	+5°C	+40°C
Temperature Change Rate	-10°C/hr	+10°C/hr
Pressure (10,000' max altitude)	$6.9 \times 10^4 \text{ N/m}^2$ (520 Torr)	$1 \times 10^5 \text{ N/m}^2$ (760 Torr)
Relative Humidity	30%	70%

Table 3: Operating Environments for Handling, Transportation, and Storage

2.4 Point Design (Modal Payload)

Chyba and Phillips [5] proposed guidelines for future life searches: 1. Conduct experiments with contrasting life definitions if payload permits. 2. Prioritise biochemical definitions given limited payload. 3. Establish geological and chemical context. 4. Ensure life-detection experiments provide valuable information regardless of results. 5. Contextualise guidelines within exploration and discovery-driven science.

These guidelines are currently used to inform missions like Dragonfly, which will explore Titan’s prebiotic chemistry and potential habitability using a suite of scientific instruments.

Hence, the following modal payloads are selected:

1. **Organic Compositional Analyser (OCA):** Used to detect and quantify key organic molecules that might indicate past or present life.
2. **Microscope:** Designed to identify microbial cells and structures. It combines spectroscopy and atomic force microscopy (AFM) or optical light microscopy (OM) to analyse cellular properties and detect potential bio-signatures in the ice.
3. **Vibrational Spectrometer (VS):** Characterises both organic and inorganic compounds through Raman and Deep UV fluorescence spectroscopy. This instrument detects molecular compositions and provides detailed chemical analysis of the sample.

4. **Color Stereo Imagers:** Capture images of the landing site. These imagers help characterise the surface composition and identifying potential hazards or areas of interest for sampling.
5. **Seismic Package:** Measures the thickness of Europa’s ice shell and ocean by monitoring acoustic signals from cracking events in the ice. This data is crucial for understanding the moon’s geophysical properties and potential habitability.

These investigations are highly complementary, ensuring measurement redundancy. The investigations yield high-value science, even in the absence of any potential bio-signatures, by advancing our understanding of Europa’s chemistry, geology, and potential habitability.

3 Detailed Analysis

3.1 Trade Study (Baseline vs Threshold)

Instrument	Functionality (Baseline / Threshold)	Data Volume Estimate (Base- line/Thresh- old)	Power Consumption Estimate (Base- line/Thresh- old)	Sizing Volume Estimate (Base- line/Thresh- old)	Example Instruments
Organic Compositional Analyser (OCA)	Detection of organic compounds at [1/10] picomole per gram	16 Mb per sample / Slightly lower	30 W / Slightly lower	25 L / 15 L	[QMS and] GC from the Sample Analysis at Mars (SAM) suite on MSL
Microscope	Identification of microbial cells and structures as small as [0.2/1] microns	79.7 Mb per tal / Reduced	7.4 W / Reduced	8 L / 6 L	Baseline: WATSON (2020) Threshold: Phoenix Mars Lander MECA
Vibrational Spectrometer (VS)	Characterisation of molecular compositions down to parts per [1000/100] by mass	320 Mb per sample / 80 Mb per sample	17.4 W / Lower	10 L / 8 L	Baseline: SHERLOC (2020) Threshold: RLS on the ESA ExoMars 2020 rover
Color Stereo Imagers	[3-D/2-D] images of the landing site	128 Mb per tal / Significantly lower	2.5 W / Reduced	15 L / 10 L	Baseline: Mastcam-Z on Mars 2020 Threshold: Navcam on MSL
Seismic Package	Measurement of ice shell and ocean thickness / Detection of major ice cracking events	1.1 Gb per site / 1 Mb per day	10-30 W / Similar or lower	0.1 L / 0.5 L	Baseline: SP seismometer from SEIS on InSight Threshold: SISMO seismometer from the OPTIMISM instrument on the Russian Mars 96 spacecraft

Table 4: Baseline vs Threshold Trade Study

3.2 Performance Assessment

3.2.1 Vibrational Spectrometer (VS) and Microscope

The baseline instrument, SHERLOC, has been chosen as the microscope and spectrometer for the mission. SHERLOC is a deep ultraviolet (DUV) native fluorescence and resonance Raman spectrometer [25]. It assesses life-sustaining elements such as carbon (C), hydrogen (H), nitrogen (N), oxygen (O), phosphorus (P), and sulfur (S) using a DUV laser to illuminate targets, generating characteristic Raman and fluorescence spectral responses.

Raman Scattering Theory Raman scattering is an inelastic process where incident photons interact with molecular vibrations, leading to a shift in energy of the scattered photons. The energy difference (ΔE) between the incident (E_i) and scattered (E_s) photons is specific to molecular vibrations:

$$E_i = E_s + \Delta E$$

In wavenumbers, the Raman shift ($\Delta\nu$) is:

$$\Delta\nu = \left(\frac{1}{\lambda_i} - \frac{1}{\lambda_s} \right) \times 10^7 \text{ cm}^{-1}$$

where λ_i and λ_s are the wavelengths of the incident and scattered light, respectively.

SHERLOC uses a 248.6 nm DUV laser to induce Raman scattering and fluorescence in target molecules. The Raman scattering process involves energy loss from the excitation energy (40225 cm^{-1}), defining the vibrational energy of a bond with which it interacted. This allows for the classification of chemical bonds such as C-H, CN, C=O, C=C, NHx, NOx, SOx, POx, ClOx, and OH. Fluorescence involves absorption of the laser light followed by emission as molecules return to their ground state. The emitted light has a longer wavelength, providing information about electronic transitions and the molecular environment. For example, a peak at 1000 cm^{-1} indicating a benzene ring, or an amide I band at 1650 cm^{-1} indicates protein secondary structure, etc.

SHERLOC also includes WATSON, a microscopic imager capturing detailed surface features to complement spectroscopic data. This combination allows comprehensive analysis of chemical composition and physical characteristics. While MECA on the Phoenix Lander has a smaller volume of 2.95 liters and has been successfully deployed in previous missions [9], its outdated experimental methods and the mechanical complexity of its sample wheel and translation stage (SWTS) pose a higher risk of mechanical failure. Since the instruments will be housed inside a vault with built-in radiation shielding, the mass, volume, and power requirements of the body can be excluded from calculations.

Main job	Fine-scale detection of minerals, organic molecules and potential biosignatures
Mass	Turret: 3.11 kilograms Body: 1.61 kilograms
Power	Turret: 32.2 watts Body: 16.6 watts
Volume	26.0 by 20.0 by 6.7 centimeters
Data return	79.7 Mbits (raw)
Spatial resolution	2 Cameras Autofocus and Context Imager: 10.1 micrometers WATSON Camera: 15.9 micrometers 1 Laser: 100 micrometers
Field of view	Imaging: 2.3 by 1.5 centimeters Spectroscopy: 7 by 7 millimeters

Table 5: SHERLOC-WATSON Specification [21]

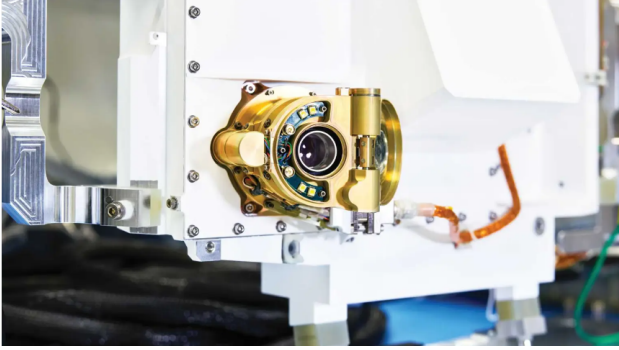


Figure 7: SHERLOC (Scanning Habitable Environments with Raman and Luminescence for Organics and Chemicals) [24]

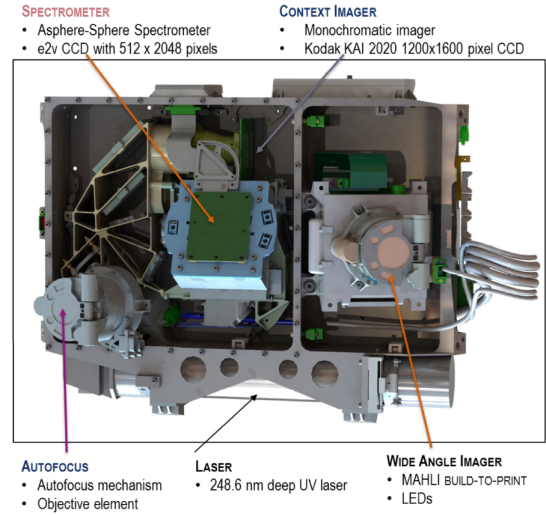


Figure 8: WATSON [24]

3.2.2 Organic Compositional Analyser

The Quadrupole Mass Spectrometer (QMS) from NASA Goddard and a 6-column Gas Chromatograph (GC) from the SAM French partners from the Sample Analysis at Mars (SAM) suite are chosen as the Organic Compositional Analyser (OCA) for the Europa Lander. The primary method for detecting organic molecules by SAM is pyrolysis, which involves heating a small sample to release gases. Each quartz cup in the sample manipulation system can hold up to 0.5 cm³ of sample, with incremental deliveries of about 0.05 cm³. The sample is heated to around 1,000°C, and as gases are released, they are carried by helium gas to the QMS for analysis. Water is typically released early, moderately volatile organics between 300 - 600 °C, and refractory organics and mineral breakdown gases at higher temperatures. For further analysis, organic molecules can be trapped and separated via GCMS. The mass range of the QMS is 2 - 535 daltons, and detection limits are approximately 10⁻¹⁴ to 10⁻¹³ moles, depending on the compound [21].

Parameter	Details
Estimated Mass	16.4 kg [33]
Sample Analysis Time	Approximately 50 minutes per sample [33]
Sample Processing	Multiple aliquots can be processed either in a single run or sequentially
Data Volume per GC-MS Scan	16 Mb [33]
Detection Limits	0.01 parts per billion by weight (ppbw) or 100 femtomoles (fmol)
Mass Range of QMS	2-535 Daltons
Analysis Methods	Direct analysis via QMS, trapping and separating organics with GC-MS

Table 6: QMS and GC from SAM Specification

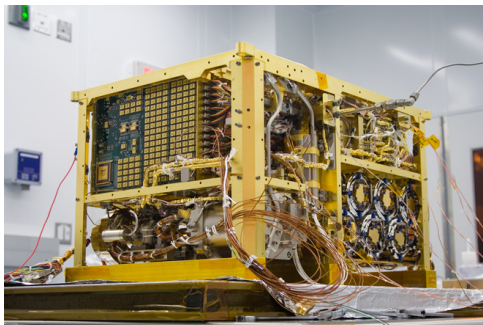


Figure 9: The SAM instrument suite, without side panels [21]

3.2.3 Color Stereo Imagers

The modal payload, Mastcam system on the Mars Science Laboratory (MSL) rover, comprises of two camera heads and two Digital Electronics Assemblies (DEAs). These cameras are mounted on the MSL Remote Sensing Mast, with the DEAs housed within a single temperature-controlled box inside the rover.



Figure 10: Mastcam-34 and 100

Parameter	Mastcam-34 & Mastcam-100 Combined
Main Job	High-definition video, panoramic color, and 3D imaging with zoom lens for distant targets.
Location	Mounted at human-eye height (2 meters), with cameras 24.2 cm apart for stereo vision.
Mass	4 kg
Power	17.4 watts
Volume	Camera Head: 11 x 12 x 26 cm, DEA: 22 x 12 x 5 cm, Calibration Target: 10 x 10 x 7 cm
Data Return	148 Mb per sol
Color Quality	Similar to consumer digital cameras (2-megapixel)
Image Size	1600 x 1200 pixels max
Resolution	450 microns pp @2 m; 22 cm pp @1 km (MC-34), 150 microns pp @2 m; 7.4 cm pp @1 km (MC-100)
Focal Length	34 mm, f/8 (Mastcam-34), 100 mm, f/10 (Mastcam-100)
Field of View	15° square (Mastcam-34), 5.1° square (Mastcam-100)
Focus	Mechanical focus and autofocus; 2.1 meters to infinity
Filters	MC-34: 440, 525, 550, 675, 750, 865, 1034nm; MC-100: 440, 525, 550, 800, 905, 935, 1035nm
Video	720p HD (1280 x 720 pixels) at 10 fps
Stereo Baseline	24.5 cm

Table 7: Mastcam Specifications

The Mastcam system's imaging capabilities, utilizing Bayer Pattern Filter CCDs, capture nearly true color images similar to human vision on Mars, eliminating the need for separate red, green, and blue images, thus reducing data volume. Each camera head features a filter wheel for various narrow visible and near-infrared wavelengths. Both cameras can record 720p HD video at approximately 10 frames per second, with an 8 Gbyte internal storage. Thumbnail images are sent to Earth for review, enabling the science team to select full frames for detailed analysis. The Mastcam quickly acquires 360° Martian landscape mosaics within an hour. While capable of stereo imaging, this is not a major focus due to differing focal lengths.

The Mastcam system is selected over ChemCam and Navcam for the Europa Lander due to its higher data rates and resolution, critical for scientific objectives. ChemCam, a laser-induced breakdown spectroscopy (LIBS) instrument, offers compositional data but with limited imaging resolution of 1000 microns per pixel at 2 meters and high data rates, requiring more bandwidth and storage. Navcam, designed for navigation, provides wide-angle, lower-resolution images (400 microns per pixel at 2 meters), unsuitable for detailed analysis. Mastcam's resolution (150 microns per pixel at 2 meters for Mastcam-100 and 450 microns per pixel for Mastcam-34), versatile imaging, and efficient data handling make it ideal for the Europa mission, providing the detailed color and 3D images needed for surface analysis and site identification within the robotic arm's workspace.

To operate on Europa, the Mastcam system will require enhancements in radiation shielding to withstand Jupiter's magnetosphere. This involves adding radiation-absorbing materials and using radiation-hardened electronics to ensure operational reliability.

3.2.4 Geophysical Sounding System

The Short Period (SP) seismometer, designed and built by Oxford University and Imperial College London, has been selected as the Geophysical Sounding System (GSS) for the Europa Lander. While the SP seismometer is also deployed in the SEIS instrument on the Mars InSight mission alongside the Very Broad Band (VBB) pendulum sensors [10], the Europa Lander does not require the VBB probes due to the differing seismic environments and scientific objectives. Mars, with its relatively low levels of seismic activity, necessitates the highly sensitive VBB probes to detect faint seismic signals across a broad frequency range. In contrast, Europa experiences a more dynamic seismic environment due to tidal forces from Jupiter, leading to significant stress and cracking on its ice shell. This results in frequent and intense seismic activity that can be effectively monitored using the less sensitive SP seismic probes [33].

One of the scientific objectives of the Europa Lander is to assess the habitability of Europa’s ice shell and subsurface ocean. The SP probes are sufficient for this purpose and offer the operational advantage of being deployable internally within the lander vault, eliminating the need for complex and risky external deployment mechanisms required by the VBB probes. This simplifies the mission design and enhances reliability in Europa’s harsh conditions. The SP sensors are about the size of a coin, with a mass of just 1 gram compared to the VBB’s 190 grams [27]. Additionally, the VBB sensors are fragile and prone to damage from the violent shocks experienced during space missions, such as lift-off and landing. The SP seismometer has been engineered to withstand these mechanical stresses and can detect vibrations in all three spatial directions. Despite its small size and apparent simplicity, the SP seismometer can detect vibrations in all three spatial directions, i.e. one vertical and two horizontal axes (though in contrast to the VBB seismometer, the pendulums are not inclined). An arrangement of electrodes enables the displacement of the mobile mass to be measured. The system has been designed so that it can continue to function even if the horizontal levelling system has failed to achieve optimal positioning.

Operating within a frequency range of 0.1 to 100 Hz, the SP seismometer can detect seismic events at various depths, enabling it to detect reflected and transformed body waves generated by larger, potentially infrequent or distant seismic events (depths of 100 to over 250 meters), as shown in Figure 12. This seismometer will differentiate these signals from the background noise produced by surface waves from more frequent, shallower events (less than 50 meters in depth). Despite its small size and simplicity, the SP seismometer’s robustness and sensitivity make it the optimal choice for the Europa Lander’s geophysical sounding system.

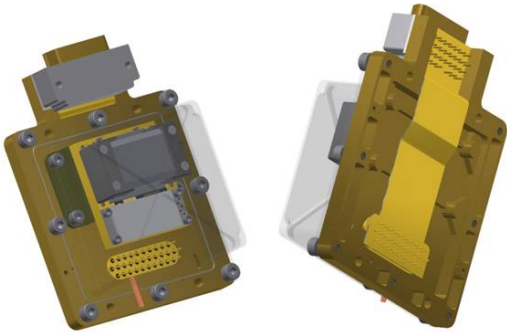


Figure 11: 3D representation of the SEIS seismometer short-period sensor

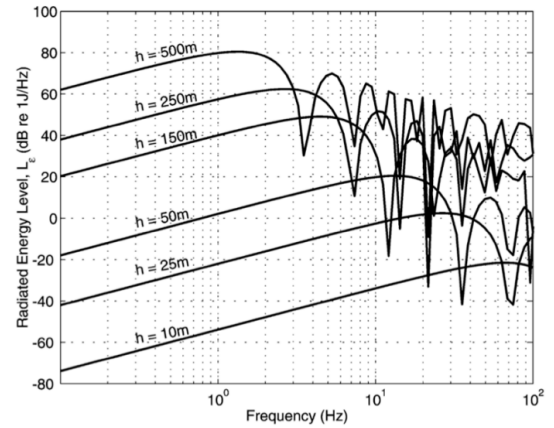


Figure 12: Energy and frequency response of simulated seismic events at depth h within Europa’s ice shell [16]

3.2.5 Vault Design

Sample Transfer System Samples on Europa cannot rely on gravity for collection or transfer, requiring a sample transfer system. Therefore, a system like CADMES will be mounted on the lander vault [17], facilitating the transfer of samples from the sampling arm to the lander. It preserves sample integrity (volume, temperature) and ensures samples remain raw and unprocessed at ± 150 K or 10 K above the collection

temperature until delivery. The system dissolves samples in water, filters them through two stages to isolate desired particles, and presents them for analysis. Each sample processing chain is assumed to include necessary ancillary hardware.

Layout The layout of the payload vault is color-coded for clarity. The blue section represents the computing systems assigned by the CD&H team, which occupy 25 liters. The yellow section indicates two computing systems for the antennas, occupying 8.6 liters. The red section marks the OCA, which occupies 25 liters, while the green section indicates the microscope and VS, occupying 3.5 liters. The CADMES (Centrifugal Automated Drill and Material Extraction System) is positioned in the direction of the robotic arm to efficiently receive and process materials.

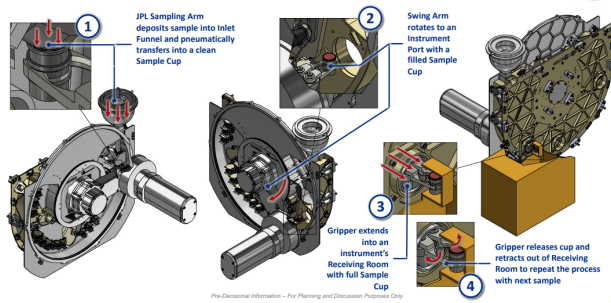


Figure 13: Collaborative Acceptance and Distribution for Measuring European Samples (CADMES) Operations

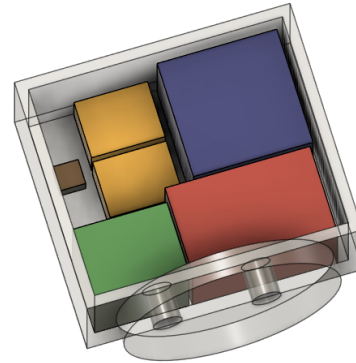


Figure 14: Vault top down view

4 Discussion

4.1 Utility Assessment

The total payload mass amounted to 69.55 kg, with a safety factor (SF) of 1.3. The mission's power usage per sampling cycle is 1117.5 Whr, allowing for the collection of two samples per site. The mission timeline is divided into "tals," a term similar to "sols" on Mars, representing an orbital period of approximately one Earth day. Each tal includes a monitoring phase and a sampling phase, with the orbiter in view of the lander for about 10 hours, facilitating data transfer and command updates.

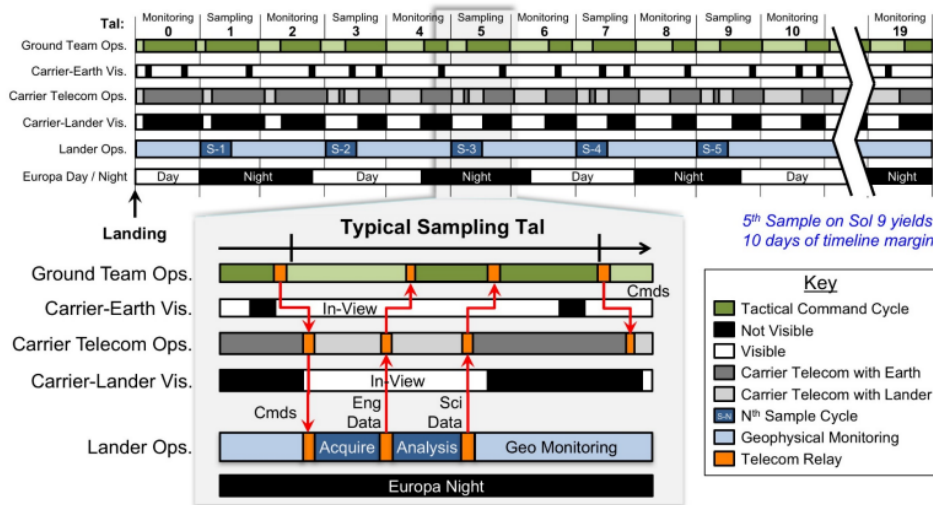


Figure 15: Mission timeline [33]

The mission plan includes four sampling tals over a 20-day period, with two samples collected at each of two sites, followed by a hopping maneuver on the fifth tal to a new site for additional sampling. This plan ensures

the collection of diverse samples, which is critical for a comprehensive assessment of Europa’s habitability. The data volume generated per sampling cycle is approximately 1.5 Gbits, leading to a total data volume of 7 Gbits, well within the 9 Gbit transfer constraint. This margin allows for additional scientific activities, such as trenching and extended analysis, providing flexibility in case of incomplete data transfer or unexpected discoveries.

The mission design adheres to the constraints specified in Table 2, demonstrating the feasibility of the proposed payload and operations within the given budget and technical limits. The selected instruments provide complementary capabilities, ensuring robust and redundant measurements critical for detecting potential biosignatures and understanding Europa’s geophysical properties.

4.2 Improvements

4.2.1 Instrument Failure Rate

A more comprehensive study on the success rate of the chosen payloads is essential to enhance the mission’s reliability. Historical data from similar missions should be analysed to identify common failure modes and success factors. For example, studies like NASA’s Sounding Rockets Annual Report [2] and ”Small-Satellite Mission Failure Rates” by Stephen A. Jacklin [12] provide valuable insights into the reliability of space instruments. Incorporating lessons learned from these studies can help mitigate risks and improve the overall success rate of the mission. Implementing rigorous testing protocols, redundancy in critical systems, and using proven technologies wherever possible can further enhance reliability.

4.2.2 Payload Mounting Options

Secure and efficient mounting of each individual payload within the vault is crucial to ensure their integrity during the mission. Several mounting options should be considered and tested to find the optimal solution. Modular mounting panels, such as a passive damping system (PDS) used in the ISS’s EXPRESS racks [6], can absorb shocks and vibrations, ensuring the payloads withstand the impacts experienced during launch and landing. Additionally, using customisable mounting brackets, shock-absorbing mounts, and modular panels can provide flexibility in the arrangement and secure attachment of payloads. Thorough testing under simulated mission conditions will help validate the effectiveness of these mounting options and ensure the payloads’ operational integrity.

4.2.3 Extended Mission Duration and Additional Objectives

While the primary mission duration is set at 30 days, extending the operational period could significantly enhance the scientific return. Developing strategies to conserve energy and manage resources effectively can help prolong the mission duration. Additionally, incorporating advanced data compression techniques and optimising data transmission protocols can maximise the data sent back to Earth, allowing for more extensive analysis. Exploring additional mission objectives, such as deploying secondary instruments or conducting in-depth geological surveys, can provide a broader understanding of Europa’s environment. Engaging with the scientific community to gather diverse perspectives and insights can help refine mission objectives and identify innovative approaches to maximise the mission’s scientific value.

4.2.4 Future Missions and Collaboration

The insights gained from this mission will be invaluable for planning future explorations of Europa and other icy moons. Collaborative efforts with international space agencies and scientific organisations can foster the development of advanced technologies and methodologies for astrobiological research. Sharing data and findings with the global scientific community can accelerate progress in the search for extraterrestrial life and enhance our understanding of the conditions that support life beyond Earth.

In conclusion, the proposed Europa lander mission presents a robust and feasible plan for exploring one of the most promising environments for extraterrestrial life in our solar system. By addressing potential improvements and leveraging lessons learned from past missions, we can enhance the mission’s success rate and scientific impact, paving the way for future explorations of Europa and beyond.

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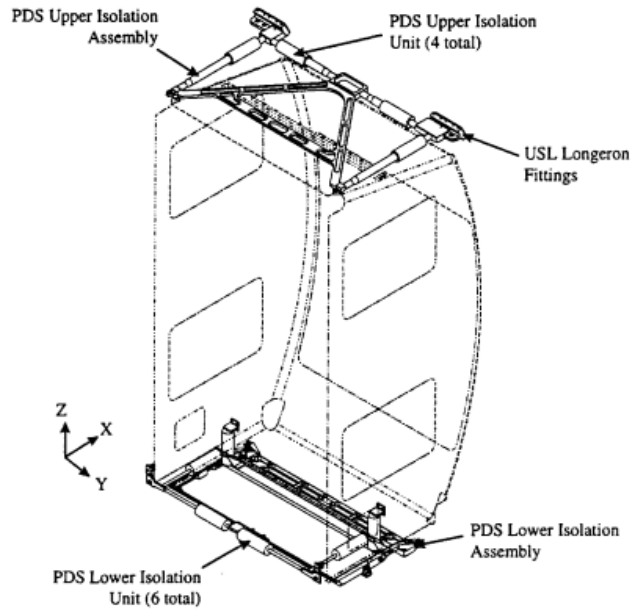


Figure 16: Illustration of PDS showing upper and lower isolation assemblies

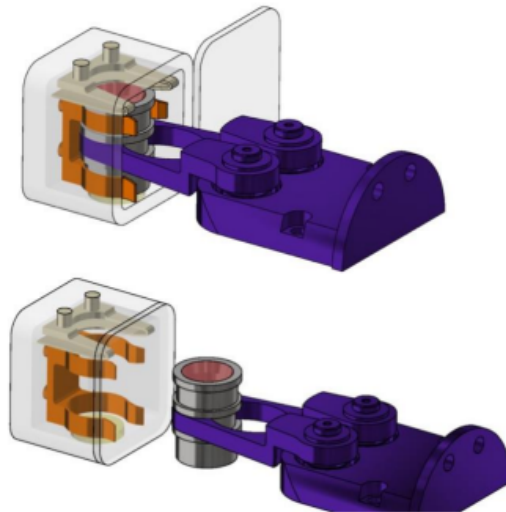


Figure 17: Generic receiving room and cup showing CADMES gripper removing a cup from an example set of interface clips

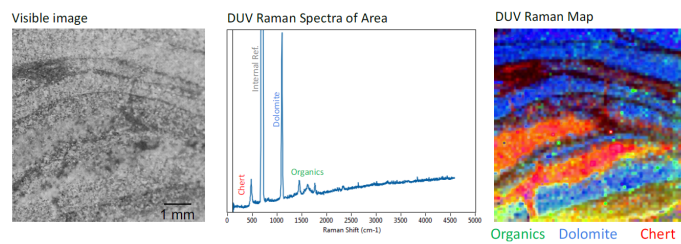


Figure 18: SHERLOC science on the Stelley Pool Stromatolite using the lab breadboard MOBIUS instrument [25]

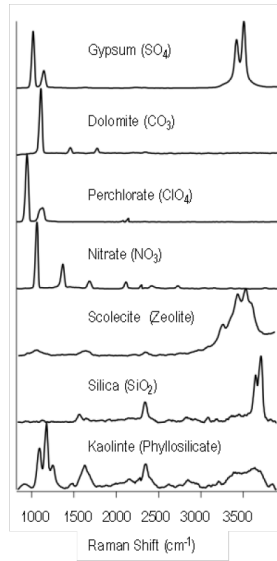


Figure 19: Raman spectra of astrobiologically relevant minerals [25]

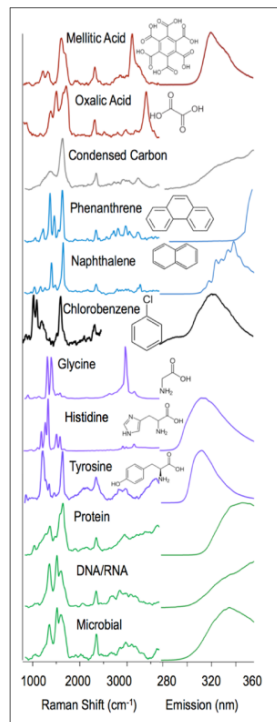


Figure 20: Raman and Fluorescence spectra of some important organic molecules [25]